Lecture 6. Nonlinear Equations

Bisection, fixed-point iteration, contraction mapping, Newton's method

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We want x with $f(x) = 0$ for nonlinear f. There is no direct method; we iterate. Root finding and optimization are closely linked: a critical point of a differentiable g is a root of $f = g'$ (or ∇g). Statistical uses include maximum likelihood, nonlinear least squares, Bayes and MAP estimation. Examples: a root of $f(x) = x^4 - \sin x$; the maximizer of $g(x) = \log x/(1+x)$, a root of $g'(x) = \{1 + x^{-1} - \log x\}/(1+x)^2$.

6.1 Bisection

If f is continuous on $[l, r]$ with $f(l)f(r) \le 0$, the intermediate value theorem gives a root in $[l, r]$. Evaluate f at the midpoint $m = (l+r)/2$: stop if $f(m) = 0$; otherwise replace r by m if $f(l)f(m) < 0$ and l by m otherwise. After n steps the bracketing interval has length $2^{-n}(r^{(0)} - l^{(0)})$, so the midpoint is within that distance of a root. Bisection is slow (one bit per step) but always works under these weak assumptions. A typical use is computing a quantile: solve $F(x) - u = 0$ for a continuous cdf F. If F itself is only available as $\int_{-\infty}^{x} f$, each evaluation is a numerical integral (Lecture 8); this is how the inverse cdf method of Lecture 9 can be used when F^{-1} has no closed form.

6.2 Fixed-point iteration

x is a **fixed point** of F if $F(x) = x$. To find a root of f, choose F with $f(x) = 0 \iff F(x) = x$ and iterate $x^{(k+1)} = F(x^{(k)})$. (The iterative methods of Lecture 5 are the case $F(x) = Rx + c$.)

Theorem 6.1 (Contraction mapping)

Let $I \subset \mathbb{R}$ be a closed interval and $F : I \to I$ with $|F(x) - F(y)| \le L|x - y|$ for all $x, y \in I$, where $0 \le L < 1$. Then F has a unique fixed point $x^\star \in I$, the iteration converges to it from every $x^{(0)} \in I$, and

$$|x^{(n)} - x^\star| \le \frac{L^n}{1-L}|x^{(1)} - x^{(0)}|.$$

Proof. $|x^{(k+1)} - x^{(k)}| = |F(x^{(k)}) - F(x^{(k-1)})| \le L|x^{(k)} - x^{(k-1)}| \le \cdots \le L^k|x^{(1)} - x^{(0)}|$. For $m > n$, by the triangle inequality,

$$|x^{(m)} - x^{(n)}| \le \sum_{k=n}^{m-1} L^k|x^{(1)} - x^{(0)}| \le \frac{L^n}{1-L}|x^{(1)} - x^{(0)}|. \qquad (*)$$

So the sequence is Cauchy and converges to some $x^\star$, which lies in I because I is closed. F is continuous, so $x^\star = \lim x^{(k+1)} = \lim F(x^{(k)}) = F(x^\star)$. If $y^\star$ were another fixed point, $|x^\star - y^\star| = |F(x^\star) - F(y^\star)| \le L|x^\star - y^\star|$ forces $x^\star = y^\star$. Letting $m \to \infty$ in $(*)$ gives the bound. ■

The geometric rate L^n is called **linear convergence**.

Proposition 6.2

If F is differentiable on $I = [a, b]$, then $|F(x) - F(y)| \le L|x - y|$ for all $x, y \in I$ if and only if $|F'(x)| \le L$ for all $x \in I$. Hence the best Lipschitz constant is $\sup_I |F'|$.

Proof. ($\Rightarrow$) For $x \in I$ and $y_n \to x$ in I, $y_n \ne x$, the difference quotients are bounded by L in absolute value, and so is their limit $F'(x)$. ($\Leftarrow$) For $x \ne y$, the mean value theorem gives c between them with $|F(x) - F(y)| = |F'(c)||x - y| \le L|x - y|$. ■

Example 6.1 (Relaxed fixed-point iteration)

To solve $f(x) = 0$, iterate $x^{(t+1)} = x^{(t)} + \alpha f(x^{(t)})$, $\alpha \neq 0$, i.e. $F(x) = x + \alpha f(x)$. By Proposition 6.2 and Theorem 6.1, it converges whenever F maps an interval I around the root into itself and $\sup_I |1 + \alpha f'(x)| < 1$. This needs f' of constant sign on I; if, say, $0 < m \le f' \le M$ there, every $\alpha \in (-2/M, 0)$ works, and $\alpha = -2/(m + M)$ gives the best constant $L = (M - m)/(M + m)$. So the free parameter α turns f into a contraction that the plain iteration $x + f(x)$ need not be, and tunes the rate. Newton's method is the adaptive choice $\alpha = -1/f'(x^{(t)})$, which makes $F'(x^\star) = 0$. If $f = g'$ for a twice differentiable g to be minimized, the iteration with $\alpha = -\gamma$ is exactly gradient descent $x^{(t+1)} = x^{(t)} - \gamma g'(x^{(t)})$ (Lecture 7), and the condition above is the familiar step-size bound $\gamma < 2/\sup g''$.

Example 6.2

For $f(x) = x^4 - \sin x$ on $(0, 1]$, both $F_1(x) = (\sin x)^{1/4}$ and $F_2(x) = \sin x / x^3$ have the root as a fixed point. On $I = [0.2, 1]$, F_1 maps I into itself and $F_1'(x) = \cos x / \{4(\sin x)^{3/4}\}$ is decreasing with $0 < F_1'(1) < F_1'(0.2) < 1$, so F_1 is a contraction and the iteration converges. F_2 maps $[0.9, 1]$ onto the larger $[0.841, 1.075]$ with $F_2'(1) < -1$: it expands, and the iteration diverges. The choice of F matters.

6.3 Newton's method

Linearizing f at the current iterate, $0 = f(x^\star) \approx f(x^{(k)}) + f'(x^{(k)})(x^\star - x^{(k)})$, suggests

$$x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}.$$

This is fixed-point iteration with $F(x) = x - f(x)/f'(x)$, for which

$$F'(x) = \frac{f(x) f''(x)}{f'(x)^2}.$$

Theorem 6.3 (Local quadratic convergence)

Let $f(x^\star) = 0$, $f'(x^\star) \neq 0$, and f be three times continuously differentiable near $x^\star$. Then for $x^{(0)}$ close enough to $x^\star$, Newton's iterates converge to $x^\star$ and

$$\lim_{k \to \infty} \frac{|x^{(k+1)} - x^\star|}{|x^{(k)} - x^\star|^2} = \frac{1}{2}|F''(x^\star)| = \frac{|f''(x^\star)|}{2|f'(x^\star)|}.$$

Proof. $F'(x^\star) = 0$ since $f(x^\star) = 0$; by continuity $|F'| \le \frac{1}{2}$ on a closed neighbourhood I of $x^\star$, which F then maps into itself, so Theorem 6.1 gives convergence. By Taylor's theorem,

$$x^{(k+1)} - x^\star = F(x^{(k)}) - F(x^\star) = F'(x^\star)(x^{(k)} - x^\star) + \tfrac{1}{2}F''(\tilde{x})(x^{(k)} - x^\star)^2 = \tfrac{1}{2}F''(\tilde{x})(x^{(k)} - x^\star)^2,$$

with $\tilde{x}$ between $x^{(k)}$ and $x^\star$. Divide and let $k \to \infty$; $F''(x^\star) = f''(x^\star)/f'(x^\star)$ follows by differentiating F'. ∎

Quadratic convergence roughly doubles the number of correct digits per step. But it is only *local*: from a poor starting point Newton's method can wander off, converge to an unintended root, or diverge (e.g. seeking a root of $\cos x$ from different starting points). Safeguards (combining with bisection, line search) are used in practice.

6.4 Systems of equations

For $f : \mathbb{R}^n \to \mathbb{R}^n$ with Jacobian $Jf(x) = (\partial f_i/\partial x_j)$, Newton's method is

$$x^{(k+1)} = x^{(k)} - [Jf(x^{(k)})]^{-1} f(x^{(k)}),$$

implemented by solving the linear system $Jf(x^{(k)})\Delta x = -f(x^{(k)})$ (e.g. by LU) at each step, never by inverting. Cheaper variants replace the Jacobian:

- modified Newton: the fixed $J = Jf(x^{(0)})$ (factor once);
- Jacobi: $\operatorname{diag} Jf(x^{(k)})$;
- Gauss–Seidel: the lower triangle of $Jf(x^{(k)})$;
- SOR: $\frac{1}{\omega}\operatorname{diag} Jf(x^{(k)})$ plus the strict lower triangle.

These converge more slowly (typically linearly) but each step is much cheaper.